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EXPLORER

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PROBLEMS

Filter (e.g. text: "**/*.ts, !**/node_modules...")

No problems have been detected in the workspace.

unit1_7.pyunit1_5.pyunit1_6.pyunit1_4.pyunit1_3.pyunit1_2.pyco3_AT_2.ipynbCO2_AT1.ipynb X

CO2_AT1.ipynb > M1 Explanation

+ Code+ MarkdownRun AllRestartClear All OutputsJupyter VariablesOutline...

```
try:
    connection = create_database()

    # Parent-table records must be inserted before child-table records.
    insert_faculty_records(connection)
    insert_course_records(connection)

    display_faculty_courses(connection)

except sqlite3.IntegrityError as error:
    print("Data integrity error:", error)

except sqlite3.Error as error:
    print("SQLite database error:", error)

finally:
    if connection:
        connection.close()
        print("\nDatabase connection closed.")

if __name__ == "__main__":
    main()
```

[6] ✓ 0.0s

...

FACULTY AND ASSIGNED COURSE DETAILS

Faculty ID	Faculty Name	Department	Course ID	Course Name	Credits
1	Dr. Rajesh Kumar	Computer Science	201	Python Programming	4
1	Dr. Rajesh Kumar	Computer Science	202	Database Management Systems	4
2	Dr. Meena Priya	Artificial Intelligence	203	Artificial Intelligence	4
2	Dr. Meena Priya	Artificial Intelligence	204	Deep Learning	4
3	Prof. Suresh Babu	Data Science	205	Data Analytics	3
3	Prof. Suresh Babu	Data Science	206	Machine Learning	4
4	Dr. Kavitha R	Mathematics	Not Assigned	No Course	-

Database connection closed.

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CO2_AT1.ipynb > M4 Explanation

+ Code+ MarkdownRun AllRestartClear All OutputsJupyter VariablesOutline

```
        Course.Course_Name,
        Course.Credits
FROM Faculty
LEFT JOIN Course
    ON Faculty.Faculty_ID = Course.Faculty_ID
ORDER BY Faculty.Faculty_ID, Course.Course_ID
"""

records = cursor.fetchall()

print("\nFACULTY AND ASSIGNED COURSE DETAILS")
print("-" * 110)
print(
    f"{'Faculty ID':<12}"
    f"{'Faculty Name':<25}"
    f"{'Department':<25}"
    f"{'Course ID':<12}"
    f"{'Course Name':<28}"
    f"{'Credits':<8}"
)
print("-" * 110)

for record in records:
    course_id = record[3] if record[3] is not None else "Not Assigned"
    course_name = record[4] if record[4] is not None else "No Course"
    credits = record[5] if record[5] is not None else "-"

    print(
        f"{record[0]:<12}"
        f"{record[1]:<25}"
        f"{record[2]:<25}"
        f"{str(course_id):<12}"
        f"{course_name:<28}"
        f"{str(credits):<8}"
    )

def main():
    connection = None

    try:
```

[6] ✓ 0.0s

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EXPLORER

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CO2_AT1.ipynb > M4 Explanation

+ Code + Markdown | Run All Restart Clear All Outputs | Jupyter Variables Outline ...

```
cursor = connection.cursor()

course_records = [
    (201, "Python Programming", 4, 1),
    (202, "Database Management Systems", 4, 1),
    (203, "Artificial Intelligence", 4, 2),
    (204, "Deep Learning", 4, 2),
    (205, "Data Analytics", 3, 3),
    (206, "Machine Learning", 4, 3)
]

cursor.executemany("""
    INSERT OR IGNORE INTO Course (
        Course_ID,
        Course_Name,
        Credits,
        Faculty_ID
    )
    VALUES (?, ?, ?, ?)
""", course_records)

connection.commit()

def display_faculty_courses(connection):
    """
    Display faculty members and their assigned courses using an SQL JOIN.

    LEFT JOIN is used so that faculty members without an assigned course
    are also displayed.
    """
    cursor = connection.cursor()

    cursor.execute("""
        SELECT
            Faculty.Faculty_ID,
            Faculty.Faculty_Name,
            Faculty.Department,
            Course.Course_ID,
            Course.Course_Name,
            Course.Credits
```

[6] ✓ 0.0s

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unit1_7.py

unit1_5.py

unit1_6.py

unit1_4.py

unit1_3.py

unit1_2.py

CO2_AT1.ipynb > M↓ Explanation

+ Code + Markdown | ▶ Run All ↺ Restart ✕ Clear All Outputs | 📄 Jupyter Variables ☰ Outline ...

```
ON UPDATE CASCADE
ON DELETE RESTRICT

)
)

connection.commit()
return connection

def insert_faculty_records(connection):
    """Insert sample faculty records."""
    cursor = connection.cursor()

    faculty_records = [
        (1, "Dr. Rajesh Kumar", "Computer Science",
         "rajesh@university.edu", "9000000001"),
        (2, "Dr. Meena Priya", "Artificial Intelligence",
         "meena@university.edu", "9000000002"),
        (3, "Prof. Suresh Babu", "Data Science",
         "suresh@university.edu", "9000000003"),
        (4, "Dr. Kavitha R", "Mathematics",
         "kavitha@university.edu", "9000000004")
    ]

    cursor.executemany("""
        INSERT OR IGNORE INTO Faculty (
            Faculty_ID,
            Faculty_Name,
            Department,
            Email,
            Phone_Number
        )
        VALUES (?, ?, ?, ?, ?)
    """, faculty_records)

    connection.commit()

def insert_course_records(connection):
    """Insert sample course records."""
    cursor = connection.cursor()
```

[6] ✓ 0.0s

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CO2_AT1.ipynb > M4 Explanation

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Course_Name TEXT Not Null

Credits INTEGER Between 1 and 6

Faculty_ID INTEGER Foreign Key

The Faculty_ID column in the Course table refers to the Faculty_ID primary key in the Faculty table. This establishes a one-to-many relationship between the two tables.

Python Program

```
import sqlite3

def create_database():
    """Create the university database and enable foreign keys."""
    connection = sqlite3.connect("course_faculty_management.db")

    # Foreign-key checking must be enabled for each SQLite connection.
    connection.execute("PRAGMA foreign_keys = ON")

    cursor = connection.cursor()

    cursor.execute("""
        CREATE TABLE IF NOT EXISTS Faculty (
            Faculty_ID INTEGER PRIMARY KEY,
            Faculty_Name TEXT NOT NULL,
            Department TEXT NOT NULL,
            Email TEXT NOT NULL UNIQUE,
            Phone_Number TEXT UNIQUE
        )
    """)

    cursor.execute("""
        CREATE TABLE IF NOT EXISTS Course (
            Course_ID INTEGER PRIMARY KEY,
            Course_Name TEXT NOT NULL,
            Credits INTEGER NOT NULL
            CHECK (Credits BETWEEN 1 AND 6),
            Faculty_ID INTEGER NOT NULL,
```

[6] ✓ 0.0s

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EXPLORER

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CO2_AT1.ipynb > M4 Explanation

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3 low-stock product(s) were updated.

UPDATED PRODUCT INVENTORY

ID	Product Name	Category	Quantity	Unit Price
101	Laptop	Electronics	15	₹55000.00
102	Wireless Mouse	Electronics	27	₹750.00
103	Notebook	Stationery	28	₹60.00
104	Office Chair	Furniture	20	₹6500.00
105	Water Bottle	Household	25	₹350.00
106	Keyboard	Electronics	12	₹1200.00

Database connection closed.

Explanation The program creates an SQLite database named inventory_management.db. The Product table co prevent negative quantities and invalid prices. Sample product records are inserted using executemany(). The Quantity < 10; The stock of each low-stock product is increased by 20 using: UPDATE Product SET Quantity =

8. Course and Faculty Management System Database Design

Two related tables are created:

Faculty Table

Attribute Data Type Constraint

Faculty_ID INTEGER Primary Key

Faculty_Name TEXT Not Null

Department TEXT Not Null

Email TEXT Unique and Not Null

Phone_Number TEXT Unique

Course Table

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Query_processing

EXPLORER

QUERY_PROCESSING

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CO2_AT1.ipynb > M4 Explanation

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```
try:
    connection = create_database()
    insert_products(connection)

    # Display products before replenishing stock.
    display_low_stock_products(connection)

    # Add 20 units to products whose stock is below 10.
    update_low_stock(connection)

    # Display inventory after updating stock.
    display_inventory(connection)

except sqlite3.Error as error:
    print("SQLite database error:", error)

finally:
    if connection:
        connection.close()
        print("\nDatabase connection closed.")

if __name__ == "__main__":
    main()
```

0.0s

PRODUCTS WITH QUANTITY LESS THAN 10

Product ID	Product Name	Quantity
105	Water Bottle	5
102	Wireless Mouse	7
103	Notebook	8

3 low-stock product(s) were updated.

UPDATED PRODUCT INVENTORY

ID	Product Name	Category	Quantity	Unit Price
101	Apple	Electronics	15	555000.00

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- PROBLEMS
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unit1_7.py unit1_5.py unit1_6.py unit1_4.py unit1_3.py unit1_2.py co3_AT_2.ipynb

CO2_AT1.ipynb > M4 Explanation

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```
def display_inventory(connection):
    """Display the complete updated inventory."""
    cursor = connection.cursor()

    cursor.execute("""
        SELECT Product_ID, Product_Name, Category, Quantity, Unit_Price
        FROM Product
        ORDER BY Product_ID
    """)

    records = cursor.fetchall()

    print("\nUPDATED PRODUCT INVENTORY")
    print("-" * 85)
    print(
        f"{'ID':<10}"
        f"{'Product Name':<25}"
        f"{'Category':<20}"
        f"{'Quantity':<12}"
        f"{'Unit Price':<15}"
    )
    print("-" * 85)

    for record in records:
        print(
            f"{record[0]:<10}"
            f"{record[1]:<25}"
            f"{record[2]:<20}"
            f"{record[3]:<12}"
            f"₹{record[4]:<14.2f}"
        )

def main():
    connection = None

    try:
        connection = create_database()
        insert_products(connection)

        # Display products before replenishing stock.
    
```

[5] ✓ 0.05

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```
SELECT Product_ID, Product_Name, Quantity
FROM Product
WHERE Quantity < 10
ORDER BY Quantity
"""

products = cursor.fetchall()

print("\nPRODUCTS WITH QUANTITY LESS THAN 10")
print("-" * 50)

if not products:
    print("No low-stock products were found.")
else:
    print(f"{'Product ID':<15}{'Product Name':<25}{'Quantity':<10}")
    print("-" * 50)

    for product in products:
        print(f"{'product[0]':<15}{'product[1]':<25}{'product[2]':<10}")

def update_low_stock(connection):
    """
    Replenish low-stock products.

    Twenty units are added to every product whose quantity is below 10.
    """
    cursor = connection.cursor()

    cursor.execute("""
        UPDATE Product
        SET Quantity = Quantity + 20
        WHERE Quantity < 10
    """)

    number_updated = cursor.rowcount
    connection.commit()

    print(f"\n{number_updated} low-stock product(s) were updated.")

def display_inventory(connection):
```

[5] ✓ 0.0s

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PROBLEMS

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CO2_AT1.ipynb > M4 Explanation

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```
        CHECK (quantity >= 0),
        Unit_Price REAL NOT NULL
        CHECK (Unit_Price > 0)
    )
)

connection.commit()
return connection

def insert_products(connection):
    """Insert sample product records."""
    cursor = connection.cursor()

    products = [
        (101, "Laptop", "Electronics", 15, 55000.00),
        (102, "Wireless Mouse", "Electronics", 7, 750.00),
        (103, "Notebook", "Stationery", 8, 60.00),
        (104, "Office Chair", "Furniture", 20, 6500.00),
        (105, "Water Bottle", "Household", 5, 350.00),
        (106, "Keyboard", "Electronics", 12, 1200.00)
    ]

    cursor.executemany("""
        INSERT OR IGNORE INTO Product (
            Product_ID,
            Product_Name,
            Category,
            Quantity,
            Unit_Price
        )
        VALUES (?, ?, ?, ?, ?)
    """, products)

    connection.commit()

def display_low_stock_products(connection):
    """Identify and display products having quantity below 10."""
    cursor = connection.cursor()

    cursor.execute("""
```

[5] ✓ 0.0s

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CO2_AT1.ipynb > M4 Explanation

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[4] ✓ 0.0s

...

Balance of account 1002 updated successfully.

CUSTOMER ACCOUNT DETAILS

A/C No.	Customer Name	Type	Balance	Phone	Email	Branch
1001	Arun Kumar	Savings	₹25000.00	9876543210	arun@gmail.com	Chennai
1002	Priya S	Current	₹52000.00	9876543211	priya@gmail.com	Coimbatore
1003	Rahul M	Savings	₹18000.00	9876543212	rahul@gmail.com	Madurai

Database connection closed.

7. Inventory Management System Database Design

The Product table stores the product and stock information of a retail store.

Attribute Data Type Constraint Product_ID INTEGER Primary Key Product_Name TEXT Not Null Category TEXT Not Null Quantity INTEGER Not Null

than zero Python Program

[9] ✓ 0.0s

```
import sqlite3

def create_database():
    """Connect to the inventory database and create the Product table."""
    connection = sqlite3.connect("inventory_management.db")
    cursor = connection.cursor()

    cursor.execute("""
        CREATE TABLE IF NOT EXISTS Product (
            Product_ID INTEGER PRIMARY KEY,
            Product_Name TEXT NOT NULL,
            Category TEXT NOT NULL,
            Quantity INTEGER NOT NULL,
            Price REAL NOT NULL
        )
    """)
```

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CO2_AT1.ipynb > M4 Explanation

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```
        f"{'Email':<25}"
        f"{'Branch':<15}"
    )
    print("-" * 110)

    for record in records:
        print(
            f"{record[0]:<10}"
            f"{record[1]:<20}"
            f"{record[2]:<12}"
            f"₹{record[3]:<13.2f}"
            f"{record[4]:<15}"
            f"{record[5]:<25}"
            f"{record[6]:<15}"
        )

def main():
    connection = None

    try:
        connection = create_database()
        insert_accounts(connection)

        # Update the balance of account number 1002.
        update_balance(connection, 1002, 52000.00)

        display_accounts(connection)

    except sqlite3.Error as error:
        print("SQLite database error:", error)

    finally:
        if connection:
            connection.close()
            print("\nDatabase connection closed.")

if __name__ == "__main__":
    main()
```

[4] ✓ 0.0s

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CO2_AT1.ipynb > M Explanation

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```
def update_balance(connection, account_number, new_balance):
    """Update the balance of the specified customer account."""
    if new_balance < 0:
        print("Balance cannot be negative.")
        return

    cursor = connection.cursor()

    cursor.execute("""
        UPDATE Account
        SET Balance = ?
        WHERE Account_Number = ?
    """, (new_balance, account_number))

    connection.commit()

    if cursor.rowcount > 0:
        print(f"\nBalance of account {account_number} updated successfully.")
    else:
        print(f"\nAccount number {account_number} was not found.")

def display_accounts(connection):
    """Display all customer account records."""
    cursor = connection.cursor()

    cursor.execute("""
        SELECT Account_Number, Customer_Name, Account_Type,
               Balance, Phone_Number, Email, Branch_Name
        FROM Account
        ORDER BY Account_Number
    """)

    records = cursor.fetchall()

    print("\nCUSTOMER ACCOUNT DETAILS")
    print("-" * 110)
    print(
        f"{'A/C No.':<10}"
        f"{'Customer Name':<20}"
        f"{'Type':<12}"
    )
```

[4] ✓ 0.0s

FileEditSelectionViewGo<=>Query_processing

EXPLORER

QUERY_PROCESSING

app.pybank_management.dbCO2_AT1.ipynbCO3_AT2.ipynbconnecting_sql_in_py.ipynbcourse_faculty_management.dbdata_cleaning.csvData_Cleaning.ipynbexample.csvinventory_management.dbmarket_data.csvmarket_data.xlsxnull_dataset.pyOsql.dbunit1_1.pyunit1_2.pyunit1_3.pyunit1_4.py

PROBLEMS

Filter (e.g. text, **/*.ts, !**/node_modules...)

No problems have been detected in the workspace.

unit1_7.pyunit1_5.pyunit1_6.pyunit1_4.pyunit1_3.pyunit1_2.pyco3_AT

CO2_AT1.ipynb > M4 Explanation

+ Code + Markdown Run All Restart Clear All Outputs Jupyter Variables Outline

```
        Email TEXT UNIQUE,
        Branch_Name TEXT NOT NULL
    )
    """
)

connection.commit()
return connection

def insert_accounts(connection):
    """Insert sample customer account details."""
    cursor = connection.cursor()

    account_records = [
        (1001, "Arun Kumar", "Savings", 25000.00,
         "9876543210", "arun@gmail.com", "Chennai"),
        (1002, "Priya S", "Current", 45000.00,
         "9876543211", "priya@gmail.com", "Coimbatore"),
        (1003, "Rahul M", "Savings", 18000.00,
         "9876543212", "rahul@gmail.com", "Madurai")
    ]

    cursor.executemany("""
        INSERT OR IGNORE INTO Account (
            Account_Number,
            Customer_Name,
            Account_Type,
            Balance,
            Phone_Number,
            Email,
            Branch_Name
        )
        VALUES (?, ?, ?, ?, ?, ?, ?)
    """, account_records)

    connection.commit()

def update_balance(connection, account_number, new_balance):
    """Update the balance of the specified customer account."""
    if new_balance < 0:
```

[4] ✓ 0.0s

Sakthi postgres

FileEditSelectionViewGo...<-->Query_processing

EXPLORER

QUERY_PROCESSING

app.pybank_management.dbCO2_AT1.ipynbCO3_AT2.ipynbconnecting_sql_in_py.ipynbcourse_faculty_management.dbdata_cleaning.csvData_Cleaning.ipynbexample.csvinventory_management.dbmarket_data.csvmarket_data.xlsxnull_dataset.pyOsql.dbunit1_1.pyunit1_2.pyunit1_3.pyunit1_4.py

PROBLEMS

Filter (e.g. text, **/*.ts, !**/node_modules...)

No problems have been detected in the workspace.

unit1_7.pyunit1_5.pyunit1_6.pyunit1_4.pyunit1_3.pyunit1_2.pyco3_AT2.ipynb

CO2_AT1.ipynb > M4 Explanation

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+ Code + Markdown

6. Banking Account Management System Database Design

The Account table stores customer bank account information.

Attribute Data Type Constraint

Account_Number INTEGER Primary Key

Customer_Name TEXT Not Null

Account_Type TEXT CHECK constraint

Balance REAL Default 0, must be non-negative

Phone_Number TEXT Unique

Email TEXT Unique

Branch_Name TEXT Not Null

```
import sqlite3

def create_database():
    """Connect to the database and create the Account table."""
    connection = sqlite3.connect("bank_management.db")
    cursor = connection.cursor()

    cursor.execute("""
        CREATE TABLE IF NOT EXISTS Account (
            Account_Number INTEGER PRIMARY KEY,
            Customer_Name TEXT NOT NULL,
            Account_Type TEXT NOT NULL
                CHECK (Account_Type IN ('Savings', 'Current')),
            Balance REAL NOT NULL DEFAULT 0
                CHECK (Balance >= 0),
            Phone_Number TEXT UNIQUE,
            Email TEXT UNIQUE,
            Branch_Name TEXT NOT NULL
        )
    """)
```